

The Universal Logarithmic Singularity of the Endpoint Numerator

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Abstract

The sharp deterministic coefficient law identifies the endpoint radius but does not by itself identify the boundary singularity. Using the exact odd anchor and the exponentially decaying even trace remainder, we prove that the endpoint-scaled logarithm is exactly $\log(1 - w) + w$ plus a function analytic on a strictly larger disk. A repository-certified block constant gives an explicit lower radius $1.0376199\dots$ for that remainder. This is an all-order deterministic theorem, not convergence of a noisy spectrum.

1 Normalized anchors

Put $q_* = (r_H \lambda)^{-1}$, $\rho_* = q_*^{-1}$, and $b_n = a_n \rho_*^n$. The odd formula of RH-263 gives

$$b_n = (1 + \lambda^{-n})^{-1} \quad (n \geq 3 \text{ odd}).$$

For $n = 2m$, RH-268 writes

$$b_{2m} = 2\lambda^{2m} \operatorname{tr}(T^m) + \frac{1 - 2\lambda^{-m}}{1 - \lambda^{-2m}}.$$

If $m = 3k + j$, the certified trace estimate is

$$|\lambda^{2m} \operatorname{tr}(T^m)| \leq C_j \delta^k, \quad \delta < 0.801254.$$

2 Singularity theorem

Theorem 2.1. *There is a function H_{reg} analytic for*

$$|w| < r_{\text{reg}}, \quad r_{\text{reg}} \geq 0.801254^{-1/6} = 1.0376199142\dots,$$

such that

$$-\sum_{n \geq 2} \frac{a_n \rho_*^n}{n} w^n = \log(1 - w) + w + H_{\text{reg}}(w).$$

Proof. Write $b_n = 1 + r_n$. At odd orders, $r_n = O(\lambda^{-n})$. The explicit even endpoint fraction differs from one by $O(\lambda^{-m}) = O(\lambda^{-n/2})$. The trace term is $O(\delta^{n/6})$ after separating the three residue classes of m . Consequently $\sum r_n w^n / n$ converges normally on every compact subdisk of radius $\min\{\lambda, \sqrt{\lambda}, \delta^{-1/6}\} > 1$. Finally,

$$-\sum_{n \geq 2} \frac{w^n}{n} = \log(1 - w) + w,$$

which proves the claim. □

Corollary 2.2. *The logarithmic branch point at $w = 1$ is the complete nonanalytic endpoint part of the deterministic numerator logarithm. In particular, subtracting $\log(1 - w) + w$ strictly enlarges the analytic disk.*

3 Scope

The result uses exact all-order coefficient identities and a certified geometric trace remainder; no finite-order fit is promoted. It neither constructs a positive or integer spectral cloud nor proves endpoint H^2 convergence of the actual mismatch. Gates A–E remain false/open.